

# Apurva Umredkar

[apoorv.umredkar@outlook.com](mailto:apoorv.umredkar@outlook.com) | +1-716-750-8918 | Buffalo, NY | [LinkedIn](#) | [GitHub](#)

## SUMMARY

Graduate student in Computer Science with 2 years of enterprise software engineering experience and deep specialization in applied AI/ML, including generative and agentic AI. Proficient in building scalable machine learning systems from prototype to deployment, with strong technical communication and cross-functional collaboration skills.

## EXPERIENCE

### AI Product Development Intern, Vaspian LLC

09/2025 – Present

- Gained practical experience in integrating MCP servers, deploying containers with Docker, and managing version control using Git.
- Developed a RAG-powered FAQ chatbot using n8n workflows and Bluemesh AI Studio, leveraging prompt engineering, structured output generation, and a knowledge base built from call transcripts.
- Created a conversational AI agent in Eleven Labs and integrated it with Airtable using webhooks to automate the management of webinar invitations and customer response tracking.

### SAP ABAP Technical Consultant, Capgemini Technology Services India Limited

08/2022 – 07/2024

- Integrated external systems with SAP BRIM/SOM, serving over 10,000 users and reducing order processing time by 40% through scalable REST APIs using SAP ABAP, ABAP CDS, and BRF+.
- Collaborated with the data migration team to map over 500 fields across 5 production systems and led system integration tests for more than 25 interfaces alongside the security and QA teams.
- Resolved an average of 8-10 ServiceNow tickets weekly, encompassing production bug analysis and feature implementation through technical investigation and code development in SAP ACM.

## EDUCATION

- Master's, Computer Science (AI/ML), University at Buffalo (GPA: 3.78/4)
- Bachelor's, Electronics & Communication, National Institute of Technology Nagpur

08/2024 – Present

07/2018 – 05/2022

## PROJECTS

### Context Aware Small Intelligence on Edge

- Developed a production-ready, serverless Discord bot platform on Cloudflare's edge network, achieving sub-50ms global response times and eliminating server management overhead.
- Engineered a unified architecture integrating AI-powered features (conversational memory, web search summarization, PDF analysis) with seamless media control for Spotify and a local media library.

### UBAssist: RAG-powered University Chatbot LLMs, RAG, Cursor, Hugging Face, Docker, FAISS, LoRA, QLoRA

- Developed an end-to-end AI chatbot using a fine-tuned Microsoft Phi 1.5 model with parameter-efficient training techniques, achieving 48% faster training and 12 percentage point improvement in answer accuracy.
- Built knowledge base of 18,000+ university documents with hybrid search capabilities, enabling real-time responses while reducing incorrect answers by 30% compared to baseline models.
- Deployed scalable web application using Flask, FastAPI, and Docker on cloud infrastructure, optimizing model to run efficiently with only 13M parameters while maintaining performance.

### Grayscale to RGB Image Colorization CNN, UNet, Vision Transformers, GANs, Gradio

- Built complete machine learning workflow using U-Net architecture augmented with Vision Transformer (ViT) blocks, attention gates, and patch discriminator-based adversarial training, achieving 94.15% structural similarity with ground truth images on test set of 5,000 images.
- Improved model color fidelity using a weighted loss function for rare color classes and patch-level GAN loss to enforce realism and texture consistency.
- Deployed as an interactive Gradio application on Hugging Face Spaces, enabling real-time image colorization with sub-second inference time.

## PUBLICATIONS

“Improved search and rescue optimization based load balancing scheme of CCTV footage on edge computing environment” *Concurrency and Computation: Practice and Experience*, 2023. [DOI](#)

- Conducted Monte Carlo simulations to analyze the impact of video and network parameters on YOLO CNN-based edge video processing latency, emulating multi-device setups using ngrok tunneling.

## SKILLS

**Programming & Data:** Python, C++, SQL, NumPy, Pandas, Matplotlib, Seaborn, Plotly, PostgreSQL, MySQL, SQLite

**Machine Learning & AI:** PyTorch, TensorFlow, Keras, Scikit-learn, FAISS, ChromaDB, Qdrant, Hugging Face, Ollama, Context Engineering, n8n, MCP Servers, Google ADK

**Backend & DevOps:** FastAPI, Docker, Git, GitHub Actions